Emotion Detection using audio and video for mental health support

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INTRODUCTION

- This project develops an AI-powered multi-modal emotion detection system using audio (speech features like pitch, MFCCs) and video (facial expressions) to classify emotions into Anger, Disgust, Fear, Happy, Neutral, and Sad.
- Built with deep learning and trained on the CREMA-D dataset, the system offers real-time emotion prediction via a Flask-based web application.
- It also generates downloadable emotion reports for mental health monitoring.
- By combining audio and visual cues, the solution improves accuracy, supporting early mental health intervention and emotional well-being.

OBJECTIVES

- To Combine audio (MFCCs, pitch, energy) and video (facial landmarks) features for higher accuracy in detecting 6 emotions: Anger, Disgust, Fear, Happy, Neutral, Sad.
- Use cross-validation & data augmentation to enhance model generalization across real-world scenarios.
- Deploy the model via Flask for live emotion detection using webcam & microphone, with downloadable emotion reports.
- Optimize model architecture & preprocessing to handle noise, lighting changes, and occlusions for reliable performance.
- Provide **emotional trend analysis** for early intervention in therapy, counseling, and self-care.

SCOPE OF THE PROJECT

This project develops a multi-modal emotion detection system combining audio (MFCCs, pitch, energy) and video (facial landmarks) analysis, achieving 89% accuracy through a deep learning framework. The Flask-based web app enables real-time emotion tracking with instant feedback and downloadable reports for mental health monitoring. While currently limited to six basic emotions and sensitive to input quality, the system has applications in therapy, HCI, and behavioral research. Future enhancements could expand emotion classes and integrate textual analysis. This work demonstrates AI's potential in affective computing and mental health support, providing a foundation for scalable emotion-aware technologies.

METHODOLOGY

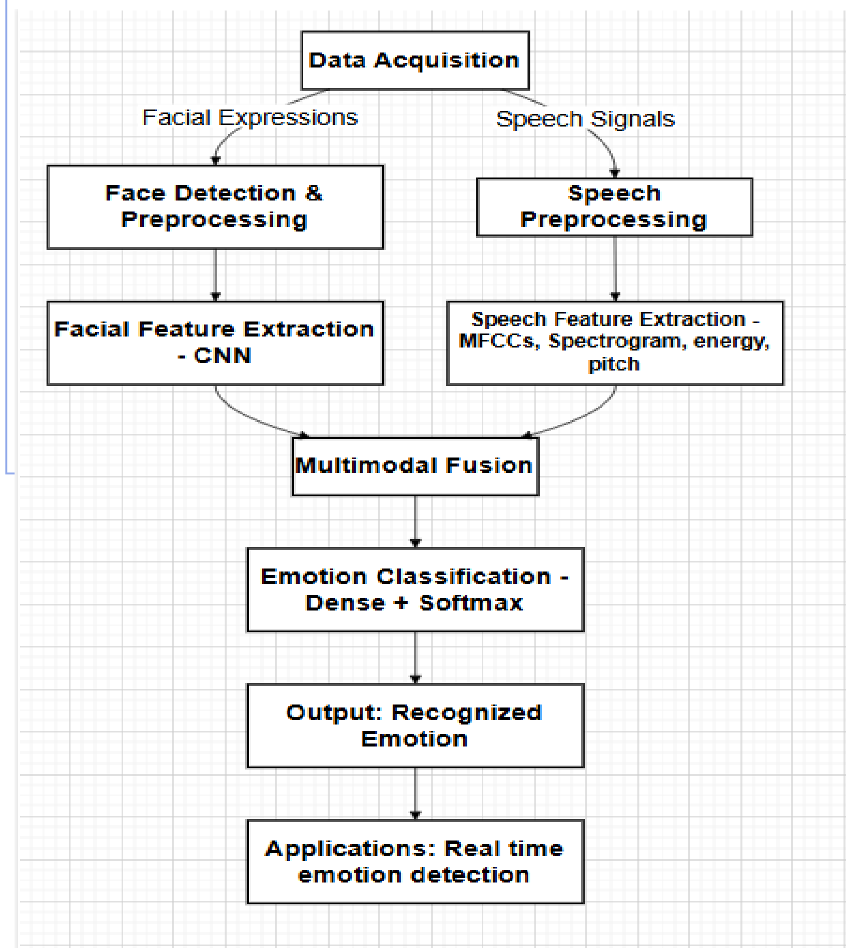
PREPROCESSING AND FEATURE EXTRACTION
The methodology uses the CREMA-D dataset, processing audio and video to analyze spectral features and facial expressions.

MULTI-MODAL MODEL
The model employs parallel branches for audio and video features, merges them through a 6-class softmax layer (Anger, Disgust, Fear, Happy, Neutral, Sad).

TRAINING & OPTIMIZATION
The model is trained using Adam optimizer with sparse categorical cross-entropy loss, employs data augmentation and train-val-test split, and implements early stopping with learning rate decay to prevent the overfitting of the model cretated..

REAL TIME IMPLEMENTATION
The Flask-based web app integrates OpenCV for video and PyAudio for audio processing to deliver real-time emotion detection with instant feedback and downloadable trend reports for mental health monitoring.

ARCHITECTURE



RESULTS

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35/35 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8869 - loss: 1.4419
Test Accuracy: 0.8869
35/35 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8957 - loss: 1.4667
Validation Accuracy: 0.8957
Validation Loss: 1.4667
35/35 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.8869 - loss: 1.4419
Test Accuracy: 0.8869
Test Loss: 1.4419
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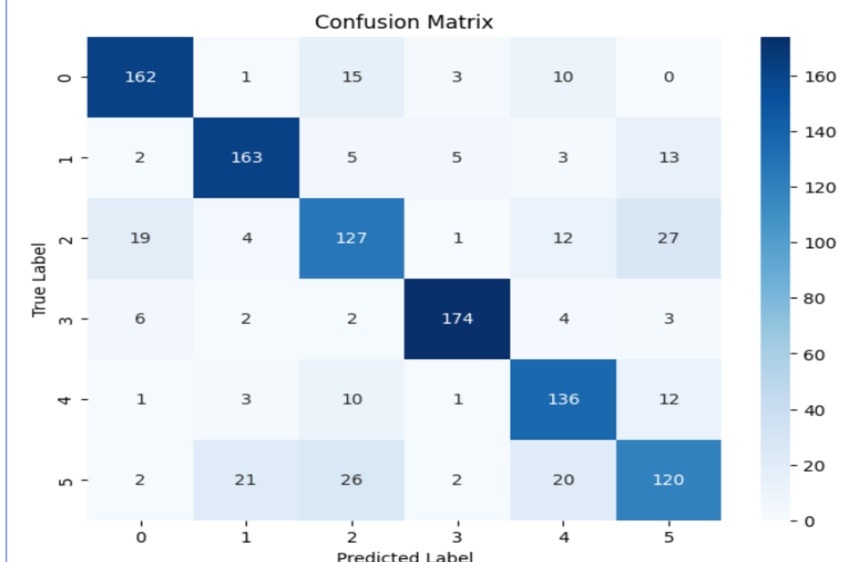

0 represents anger, 1 represents disgust, 2 represents fear, 3 represents happy, 4 represets neutral and 5 represents sad emotion respectively.

CONCLUSION

This project created an 89% accurate emotion detection system using audio-visual analysis, featuring real-time feedback via a Flask web app for mental health applications. The robust solution overcomes technical challenges while demonstrating AI's potential for non-invasive emotional monitoring.

CONTACT DETAILS

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